

Trading the Corn–Soybean Spread: Cointegration, Machine Learning and Regime Switching

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Tilburg Investment Club, Spring 2026

September 2026

Abstract

Corn and soybean compete for the same farmland and respond to the same demand and weather shocks, so a hedged position in the two should mean-revert. We test this idea with models of increasing complexity: z-score rules on the cointegrated pair and on baskets including wheat and cotton, a portfolio of pairs with Ornstein–Uhlenbeck-based capital allocation, a triple-barrier machine-learning classifier with weather features, and a Bayesian regime-switching model in which weather drives the regime. With every parameter chosen on training data and P&L measured as tradeable, capital-normalised returns, no strategy achieves an out-of-sample Sharpe ratio that is statistically distinguishable from zero. The most useful output of the project is a set of lessons on how easily a backtest of a slow, persistent spread overstates performance.

1 Introduction

Corn and soybean prices move together: farmers rotate the two crops on the same land and both are driven by feed and energy demand. If the two are *cointegrated*, a suitably hedged spread is stationary and can be bought when it is unusually low and sold when it is unusually high (Chan, 2013). A second idea was that weather in the main growing regions (the US Corn Belt and Brazil) moves the relative supply of the two crops and could help to time such trades.

The report follows the project from the simplest to the most complex model; each section corresponds to a notebook of the accompanying code, shown as [notebooks/0x] (functions in `src/`): data [notebooks/01], the corn–soybean pair [notebooks/02], wheat and cotton [notebooks/03], machine learning [notebooks/04, 05] and regime switching [notebooks/06]. Section 7 summarises the results and what we learned.

Common conventions. Strategies hold +1, 0 or −1 units of a spread (the ML strategies may also size by predicted probability). The hedge (units of each asset per unit of spread) is *frozen* when a trade is opened: letting a rolling hedge ratio drift inside a trade produces P&L that could only be earned by re-hedging every day (López de Prado, 2018, §2.4). Daily returns are P&L divided by the gross capital committed at entry, $\sum_i |u_i| P_{i,t_0}$, and Sharpe ratios are annualised with a zero risk-free rate. Every threshold, window and hyperparameter is chosen on a training sample and evaluated once on a later test sample.

2 Data

Prices. We use two price sources from Yahoo Finance. Front-month *futures* (ZC=F, ZS=F) cover 2005–2026, but the continuous series jumps at every contract roll; we realised this after the first strategies were built on it. From then on we used commodity *ETFs*, which roll internally: Teucrium CORN, SOYB and WEAT and the WisdomTree cotton ETC COTN (available from 2011; the originally used iPath BAL was delisted in 2023). Section 3 still uses futures because of their longer history; its results carry this caveat.

Table 1: Out-of-sample performance of all strategies (annualised; returns on capital; no transaction costs). Train = training sample (Sections 3–4) or walk-forward CV accuracy (Section 5); every parameter was chosen on it. SR = annualised Sharpe ratio (mean/std of daily returns $\times \sqrt{252}$); Return p.a. = mean daily return $\times 252$ (arithmetic, not compounded); Max DD = largest peak-to-trough fall of the compounded equity curve. Daily returns are P&L divided by the capital committed at entry (zero when flat); see López de Prado (2018, ch. 14) for these backtest statistics. Test years: 6.3 (Section 3), 4.4 (Section 4), 1.5–1.7 (Section 5). Full tables in [report/tables/](#).

Strategy		Train SR	Test SR	Return p.a.	Max DD
<i>Corn–soybean futures, test Oct 2019–Feb 2026</i> [notebooks/02]					
A	price levels, rolling OLS	−0.01	−0.27	−1.3%	−24%
B1	log prices, static Johansen	0.16	0.58	6.1%	−14%
B2	log prices, Kalman	0.64	0.01	0.1%	−22%
B3	log prices, rolling Johansen	0.40	0.10	1.0%	−20%
<i>Soy, corn, wheat, cotton ETFs, test Nov 2021–Apr 2026</i> [notebooks/03]					
C	rolling Johansen, all 4 commodities	0.95	0.35	2.8%	−12%
C	rolling Johansen, four 3-commodity baskets	0.14–0.98	−0.65–0.15		
D	4 commodities in logs (static, Kalman, rolling)	0.63–0.81	−0.28–−0.03		
E	pairs portfolio, equal weight	0.38	0.42	3.5%	−15%
E	pairs portfolio, MRB	0.07	0.38	3.2%	−16%
E	pairs portfolio, MRR	0.24	0.46	3.7%	−15%
<i>Machine learning, 375-day holdout, Dec 2023–Sep 2025</i> [notebooks/04, 05]					
ML	XGBoost corn–soybean, probability sizing	CV 55%	0.76	2.4%	−3%
ML	XGBoost corn–soybean, sign sizing	CV 55%	0.41	3.0%	−5%
ML	six-pair portfolio, equal weight	–	0.51	1.3%	−3%
ML	six-pair portfolio, MRB	–	0.68	2.9%	−4%
ML	six-pair portfolio, MRR	–	0.66	2.0%	−3%

Weather. Daily weather from the Open-Meteo archive for three locations in each growing region: the US Corn Belt and southern/central Brazil (corn and soybean), the US wheat belt, and China and India (cotton). For each location we keep 15 crop-stress variables (temperature, precipitation, soil moisture, evapotranspiration, humidity, wind. . .) and form 30-day backward-looking aggregates: sums for precipitation, minima for soil moisture, maxima for wind and means otherwise. Weather is forward-filled onto trading days, so every day only sees past weather.

Samples. Sections 3–4 split each sample chronologically into 70% training and 30% test. The machine-learning sections keep the last 375 labelled observations (≈ 1.5 years) as a holdout and use walk-forward cross-validation on the rest.

3 The corn–soybean pair

Tests [notebooks/02]. On the training sample (2005–2019) both prices are $I(1)$. The evidence for cointegration is weaker than it first appeared: with the correct residual-based critical values the Engle–Granger test (Engle and Granger, 1987) does *not* reject the null of no cointegration for price levels ($t = -2.88$, $p = 0.14$)¹, while the Johansen trace test (Johansen, 1991) on log prices rejects at 1% (trace 25.1 vs. 19.9). In an error-correction model soybean is pulled back towards the long-run relation ($\gamma = -0.008$, $p < 0.001$), but slowly: the half-life of the spread is 84 trading days in levels and 98 in logs, and after the training sample the static spread drifts away from zero.

Strategies. All strategies enter when the z-score of the spread relative to its trailing mean exceeds a threshold and exit when it crosses zero.

A Price levels, rolling OLS. Soybean is regressed on corn over the previous 252 days; the spread and its z-score use the same window.

¹The first version of the project used ordinary ADF p -values on the estimated residual, which gave $p = 0.003$.

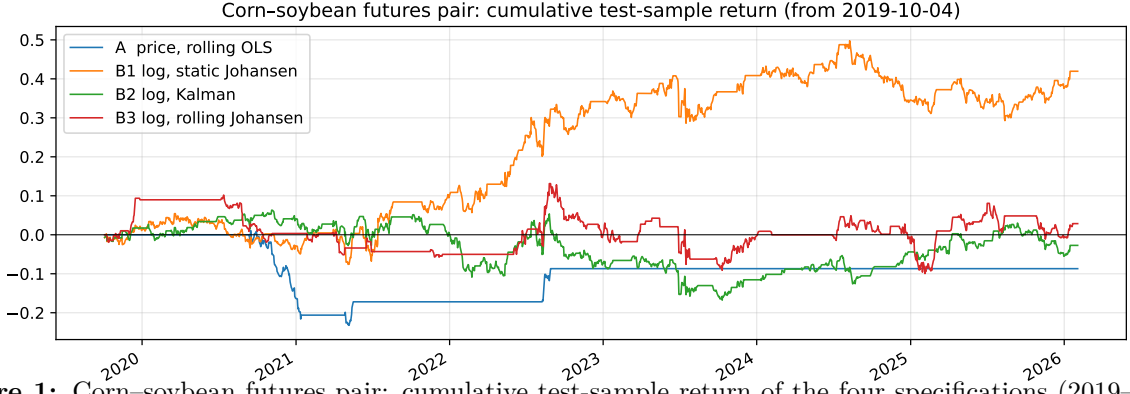


Figure 1: Corn–soybean futures pair: cumulative test-sample return of the four specifications (2019–2026).
[notebooks/02]

B Log prices. The spread $\log S_t - \beta_t \log C_t - \alpha_t$ is a portfolio of \$1 of soybean and β_t of corn, so the hedge is converted from dollars to units before computing P&L. We compare three estimators of β_t : the static Johansen vector (B1), a Kalman filter with random-walk coefficients (Chan, 2013, ch. 3) (B2), and a rolling Johansen vector (B3, traded only when yesterday’s $|\beta| \leq 2$). The z-score window equals the half-life.

Results. Table 1 (top) and Figure 1. Strategy A loses money in the training sample for every entry threshold once P&L is computed with the frozen hedge, and it loses out of sample. Among the log-price strategies, the most flexible one (Kalman) has the best training Sharpe ratio (0.64) and a test Sharpe of 0.01 — a textbook case of fitting noise. The static Johansen spread does best out of sample (Sharpe 0.58, +42% over six years), but its training Sharpe was only 0.16, and a Sharpe of 0.58 over 6.3 years has a standard error of about 0.4 (Lo, 2002): it is the best of four specifications, not evidence of an edge.

4 More commodities: baskets and a portfolio of pairs

A single spread is exposed to breakdowns of one relation, so we added wheat and cotton (ETFs, 2011–2026) [notebooks/03].

Cointegrated baskets. Johansen tests on the training sample (2011–2021) find cointegration at 95% for three of the five baskets in price levels (all four commodities; soybean–corn–wheat; corn–wheat–cotton) and for none in log prices. Where it holds, the half-life of the basket is about 35 days, half that of the pair. We trade every basket with a rolling (252-day) Johansen eigenvector in price levels (strategy C) and the four-commodity basket in log prices with the three hedge estimators of Section 3 (strategy D). Out of sample the results scatter around zero (test Sharpe from -0.65 to 0.35), with no relation to the cointegration tests: the two best baskets include one that fails the test, and all three log-price variants lose money despite training Sharpe ratios of 0.6 – 0.8 .

Portfolio of pairs (strategy E). Following Lee et al. (2023), all six pairs are traded with a z-score rule on a rolling-OLS spread, and at every rebalancing date capital is re-allocated from an Ornstein–Uhlenbeck fit to each spread: equal weights (EW), weights $\propto$ normalised reversion speed $\times$ OU log-likelihood (MRB), or fixed weights by the rank of that score (MRR). All parameters are tuned jointly on the training sample. This is the most consistent strategy of the project: a test Sharpe of 0.38 – 0.46 over 4.4 years for all three allocations (Table 1), with a similar training Sharpe for EW. The OU allocation does not improve on equal weights: all four markets share the same drivers, so the spreads are correlated and the cross-section of OU scores carries little information. An earlier version reported a Sharpe of 1.6 for this strategy; that number measured P&L in units of spread standard deviations rather than capital, and was obtained with parameters tuned on the full sample.

5 Machine learning

Set-up [notebooks/04]. Instead of a fixed rule, a classifier predicts whether buying the spread *today* ends in profit, following López de Prado (2018) (AFML). The hedge ratio of soybean on corn (ETFs) comes from a 504-day rolling OLS (labels) and a three-state Kalman filter (features): the usual two-state filter absorbs a temporarily wide spread into the hedge ratio, so we add the spread as an AR(1) hidden state, $\xi_t = [\alpha_t, \gamma_t, s_t]'$ with transition $\text{diag}(1, 1, \phi)$.

- *Labels* (AFML ch. 3–4): for an entry on every day, the P&L of a long-spread position with the hedge frozen at entry is followed until it reaches ± 2.5 daily standard deviations or 200 trading days; the label is its sign. Labels overlap heavily (mean uniqueness 0.09), so observations are weighted by their average uniqueness.
- *Features*: four Kalman features (spread level, innovation and two z-scores), spread volatility, month, weekday and 90 rolling weather variables from the US and Brazil.
- *Validation* (ch. 7, 11): the last 375 observations (Jan 2024–Jul 2025) are a holdout used once; on the rest we use walk-forward cross-validation with purging, because symmetric purged k -fold would train on later folds whose rolling-OLS and Kalman features were estimated from test-fold prices.
- *Selection and models* (ch. 4, 6, 8, 9): union of the top-20 features by MDI and by MDA, minus features with negative MDA (32 remain); XGBoost and a random forest on sequentially bootstrapped bags; 525 XGBoost and 27 pipeline configurations tuned by walk-forward log-loss.

Results. Weather carries a little information: walk-forward accuracy rises from 50% with the six non-weather features to 55% with weather, and the log-loss (0.671) is slightly below that of a coin flip ($\ln 2 = 0.693$). The signal is unstable, however — the first fold (2017–2020) has an accuracy below 50% — and hyperparameter tuning adds nothing. On the holdout XGBoost is right 60% of the time but has a log-loss of 0.718, worse than a coin flip: its probabilities are over-confident. The random forest (51%) does not beat always predicting the majority class (53%). Traded on the holdout, the signal earns Sharpe ratios of 0.3–0.8 depending on sizing and rebalancing, but over 1.5 years the standard error is about 0.9, the strategy is net long soybean, and the result is fragile: purging four observations at the dev/holdout boundary moved the Sharpe ratios by about 0.2.

Six pairs [notebooks/05]. Running the same pipeline for all six pairs, with the weather of each pair’s growing regions, and combining the pairs with the OU allocation of Section 4 gives holdout Sharpe ratios of 0.5–0.7 over about 1.7 years (Table 1). The gains come from the two soybean pairs with US-grown partners (corn, wheat); the corn–wheat and corn–cotton models never cross the confidence band ($|p - 0.5| > 0.15$) and do not trade, and the pairs with cotton lose. The result is also sensitive to seemingly innocuous choices: before the clean-up, with corn as the dependent leg and all pairs restricted to the common four-asset calendar, the same analysis gave Sharpe ratios between -0.2 and 0 .

6 Bayesian regime switching

Finally we asked whether the spread reverts to *different* equilibria in different growing conditions, with weather shifting the probability of each regime [notebooks/06]. The Kalman spread y_t follows

$$y_t = (1 - \phi) \mu_{s_t} + \phi y_{t-1} + \varepsilon_t, \quad z_t = \alpha_0 + w_t' \alpha + h_t' \kappa + \delta s_{t-1} + \eta_t, \quad s_t = \mathbb{I}[z_t > 0],$$

with $\mu_0 < \mu_1$, weather w_t , Fourier seasonal terms h_t and persistence $\delta > 0$. A Gibbs sampler draws the state path by forward-filtering backward-sampling (Carter and Kohn, 1994), the probit utilities as in Albert and Chib (1993), and the remaining parameters from conjugate conditionals.

The sampler does not converge (Geweke statistics far beyond ± 2), the regime means have posterior standard deviations of 1.5–3, and the regime probability merely follows an annual cycle. Two checks explain why. First, with $\phi \approx 0.996$ the regime mean enters each observation only as $(1 - \phi) \mu_{s_t}$, far below the noise. Simulating the model with *known* regimes shows the posterior standard deviation

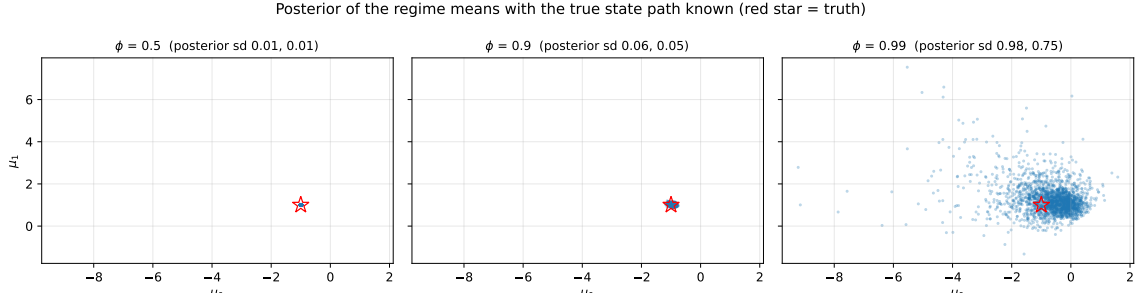


Figure 2: Posterior draws of the regime means from data simulated with known regimes ($\mu_0 = -1$, $\mu_1 = 1$, star) for three persistence levels. At the persistence of the corn–soybean spread the means are not identified. [notebooks/06]

of μ growing from 0.01 at $\phi = 0.5$ to almost 1 at $\phi = 0.99$ (Figure 2): the regime means are not identified at realistic sample sizes. Second, the bimodal histogram of the spread that motivated the model is an artefact of persistence: 87% of simulated single-regime AR(1) paths with the same ϕ and σ also prefer a two-component Gaussian mixture, and the real data sits at the 87th percentile of that null. Weather adds almost nothing to an AR(1) for daily spread changes (incremental R^2 of 0.4%, joint F -test $p = 0.06$). We therefore did not trade this model.

7 Summary

Table 1 collects the out-of-sample results. The portfolio of pairs (E) is the only strategy that performs consistently in training and test, and even its Sharpe ratio of about 0.4 over 4.4 years is within one standard error of zero (Lo, 2002). The machine-learning strategies are positive on their short holdouts, but with standard errors near 0.9 and results that move with innocuous choices. All numbers ignore transaction costs. We conclude that we did not find a robust corn–soybean (or grains) spread strategy.

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